Data Weaver - Distributed Agentic Data Pipeline - Capstone Project

Team - 2

Sprint 2

Agenda

- Team Member Roles and Responsibilities
- Problem Statement
- Project Description
- Team working agreement
- Personas
- MVP
- Technologies & Algorithms
- Diagrams
  - Architecture Diagrams
  - Context Diagram
  - Sequence Diagram
  - State Diagrams
  - Class Diagrams
- Sprint 2 Recap
- Product Backlog
- Sprint 2 Backlog
- Stories or Tasks committed for this Sprint
- Test Cases
- Metrics
  - Team velocity
  - Burndown Charts
  - Completed/Committed Ratio
- Retrospective
- Stories planned and committed for Sprint 2
- Slides App Screenshot

Team Members & Roles

Tenzing Salaka

AI Engineer - Engineer Agent

Shreya Anand Kuvalekar

Scrum Master/UI Developer/System Architect

Orges Velia

AI Engineer-Architect Agent

Priya Prajapati

Backend & Data Engineer

Kushwanth Reddy Nomula

Frontend / Dashboard Engineer

Meghana Ventrapragada

Backend & Data Engineer

Problem Statement

Organizations accumulate fragmented, inconsistent, and poorly documented datasets across departments, resulting in “data swamp” conditions that hinder analytics and decision-making. Integrating such datasets requires significant manual effort to infer schema relationships, identify primary and foreign keys, resolve type mismatches, and validate joins.Existing LLM-based data tools attempt automation but rely on probabilistic reasoning without deterministic safeguards, often hallucinating schema mappings or producing unsafe joins that corrupt downstream analytics. Additionally, most solutions lack distributed scalability and structured separation between semantic reasoning and execution logic.There is a need for a reliable, distributed, self-correcting AI system that separates semantic modeling from deterministic data execution while enforcing invariant-based validation to ensure trustworthy data integration.

Project Description

Project Name:	Data Weave
Team:	Team 2
Project Description:	For data analysts, data engineers, and organizations dealing with fragmented datasets who struggle to clean, integrate, and understand inconsistent and unstructured data sources the Distributed Agentic Data Pipeline is a multi-agent, AI-driven data integration system using a distributed architecture that automatically maps, cleans, validates, and merges datasets while generating reliable data models and documentation Unlike traditional ETL tools or LLM-based solutions that rely on manual configuration or produce unreliable, unverifiable outputs Our application uses a self-correcting two-agent architecture with deterministic validation checks to ensure accuracy, scalability, and trustworthy data transformation
Benefit Outcomes:	• Better data integration decisions with AI-assisted schema inference and validation checks • Reduced ETL troubleshooting time through automated mapping, profiling, and transformation previews • Improved trust through transparent logs, ERD visibility, and human approval checkpoints • Faster delivery of analytics-ready datasets and data dictionaries for downstream reporting
GitHub Link:	https://github.com/htmw/S2026A-Team2

Team Working Agreement

Team Principles

- Kindness, respect, and patience in all interactions
- Clear ownership and responsibilities
- Even workload distribution; break large tasks
- Document attempted solutions before escalating
- Use GitHub for all updates and reviews

Checklist to Mark Task Complete

- Code runs with no syntax/runtime errors
- Meets acceptance criteria and tests
- Peer review and multi-member verification
- Push to branch and open PR with docs updated
- All relevant documentation tagged and reviewed

Alex Martinez — Senior Data Engineer
Industry: E-commerce / Healthcare / FinTech
Experience: 5–8 years

Description

It is designed to streamline the integration of fragmented legacy data sources by automating schema interpretation and relationship mapping. It introduces deterministic validation checks—such as row-count reconciliation and null-threshold monitoring—to ensure transformation accuracy. The platform also offers end-to-end transparency into data processing steps and produces downloadable, analysis-ready datasets along with comprehensive data dictionaries.

Goals

- Consolidate multiple unstructured CSV exports originating from legacy systems
- Enable safe, accurate joins while maintaining schema consistency
- Minimize time spent on manual ETL troubleshooting
- Provide dependable, analytics-ready datasets to downstream teams

Challenges

Non-standardized column naming conventions (e.g., *cust_id*, *customerID*, *user_ref*)

- Lack of documentation for legacy data structures
- Join amplification leading to inflated or duplicated row counts
- Undetected schema mismatches corrupting downstream dashboards
- Excessive time spent validating transformations rather than developing pipelines

Priya Sharma — BI Analyst
Industry: Retail / SaaS
Experience: 3–5 years

Description

This system empowers analytics and reporting teams to independently prepare and understand integrated datasets. It provides visual entity-relationship diagrams (ERDs) to clarify how tables connect, along with transparent pipeline logs that document validation steps. Users can download analysis-ready datasets and leverage auto-generated data dictionaries to better interpret column meanings—ensuring confidence in every output produced.

Goals

- Rapidly merge datasets to support reporting needs
- Gain clear visibility into table relationships through ERD views
- Access clean, dependable data without relying on engineering teams
- Build dashboards powered by accurate, trustworthy metrics

Challenges

- Delays caused by dependence on data engineering for cleaned datasets
- Ambiguity around primary and foreign key columns
- Duplicate records distorting KPI and metric calculations
- Insufficient metadata and lack of formal data dictionaries
- Skepticism toward AI-generated joins without validation guarantees

Sprint 2

Daniel Kim — Chief Technology Officer / Head of Data

Industry: SaaS / FinTech / Healthcare

Experience: 10+ years

Description

This system is built to deliver enterprise-grade data reliability by combining deterministic validation with scalable AI infrastructure. It introduces invariant-based checks to ensure transformation accuracy, while maintaining full audit trails for governance and compliance. The architecture is designed for distributed, production-ready deployment and automatically generates documentation—including ERDs and data dictionaries—so organizations can trust, verify, and operationalize AI-driven data outputs at scale.

Goals

- Maintain consistent, high-quality data across all departments
- Lower engineering effort required for complex data integrations
- Strengthen governance, traceability, and audit readiness
- Implement scalable, AI-enabled infrastructure
- Showcase innovation through distributed AI system adoption

Challenges

- Data inconsistencies leading to conflicting reports across departments
- Significant engineering costs tied to building and maintaining ETL pipelines
- Limited schema transparency and insufficient documentation
- Concerns around AI systems generating unverified or non-auditable outputs
- Legacy or single-node infrastructure that cannot scale with enterprise demands

Minimum Viable Product

- Ingestion & Scout: Upload CSV/API via Stream lit UI; profiling, schema sniffing, PK/FK hints.
- Architect: Schema inference, ERD creation, data dictionary generation.
- Engineer: Generates Polars/Pandas code, cleaning steps, validation rules.
- Router + LLM Retry Loop: Auto-repair failed runs, regenerate improved transformations.
- Invariant Enforcement: Deterministic checks; halt + logs on violation.
- Loader & Storage: Successful pipeline saved to SQLite with logs + lineage.
- Dashboard: Stream lit UI shows ERD, profiling, invariants, logs, downloads.

Sprint 2

Technologies & Algorithms

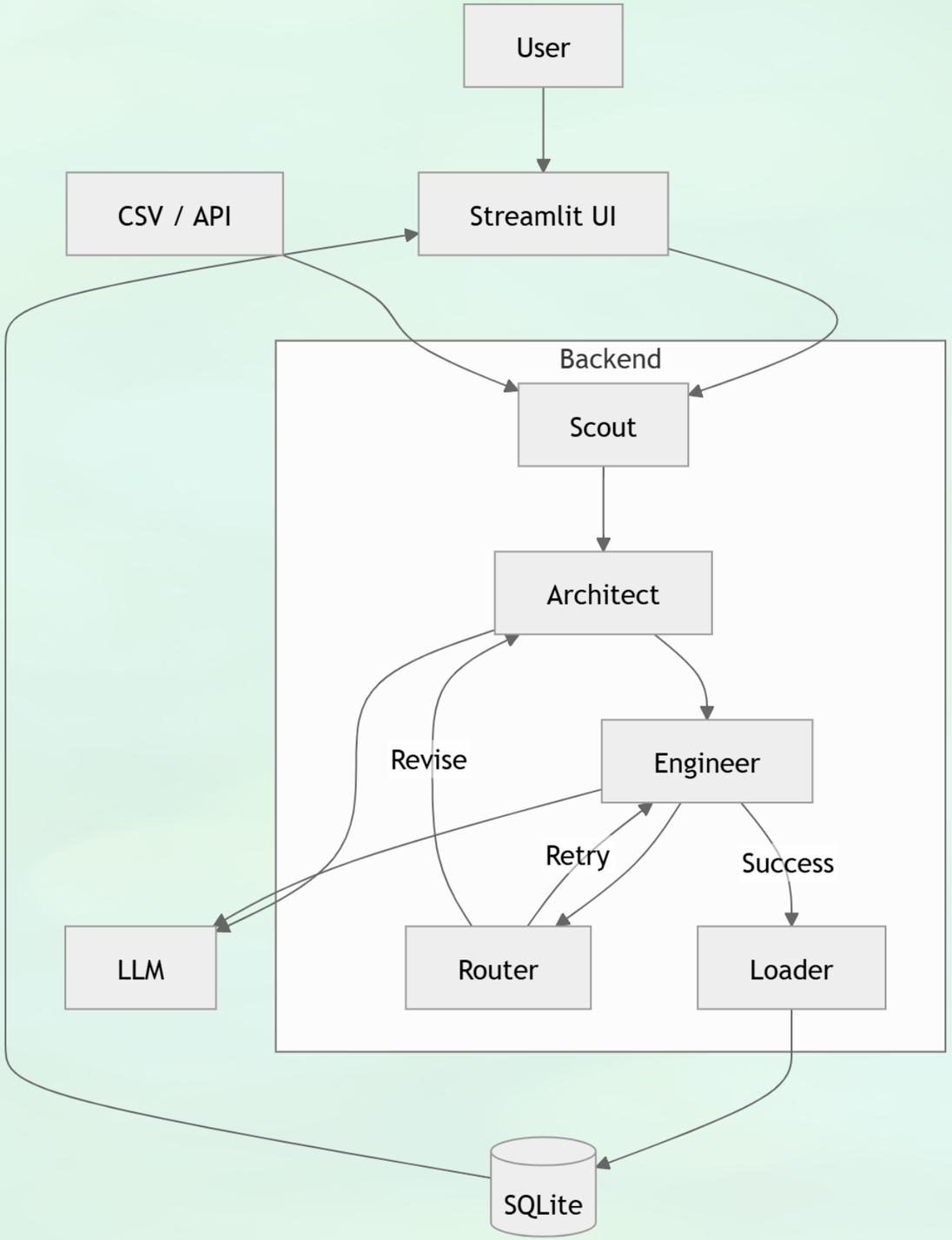
- **Architecture & Orchestration:** LangGraph — multi-agent cycles with explicit checkpoints
- API Layer: FastAPI for inter-node communication and control

- Networking: Tailscale for secure distributed connectivity

- UI: Streamlit dashboard for visualization & human checkpoints
- **Models:** local gpt-oss:20b (quantized) — Mapper high-reasoning, Aggregator high-throughput

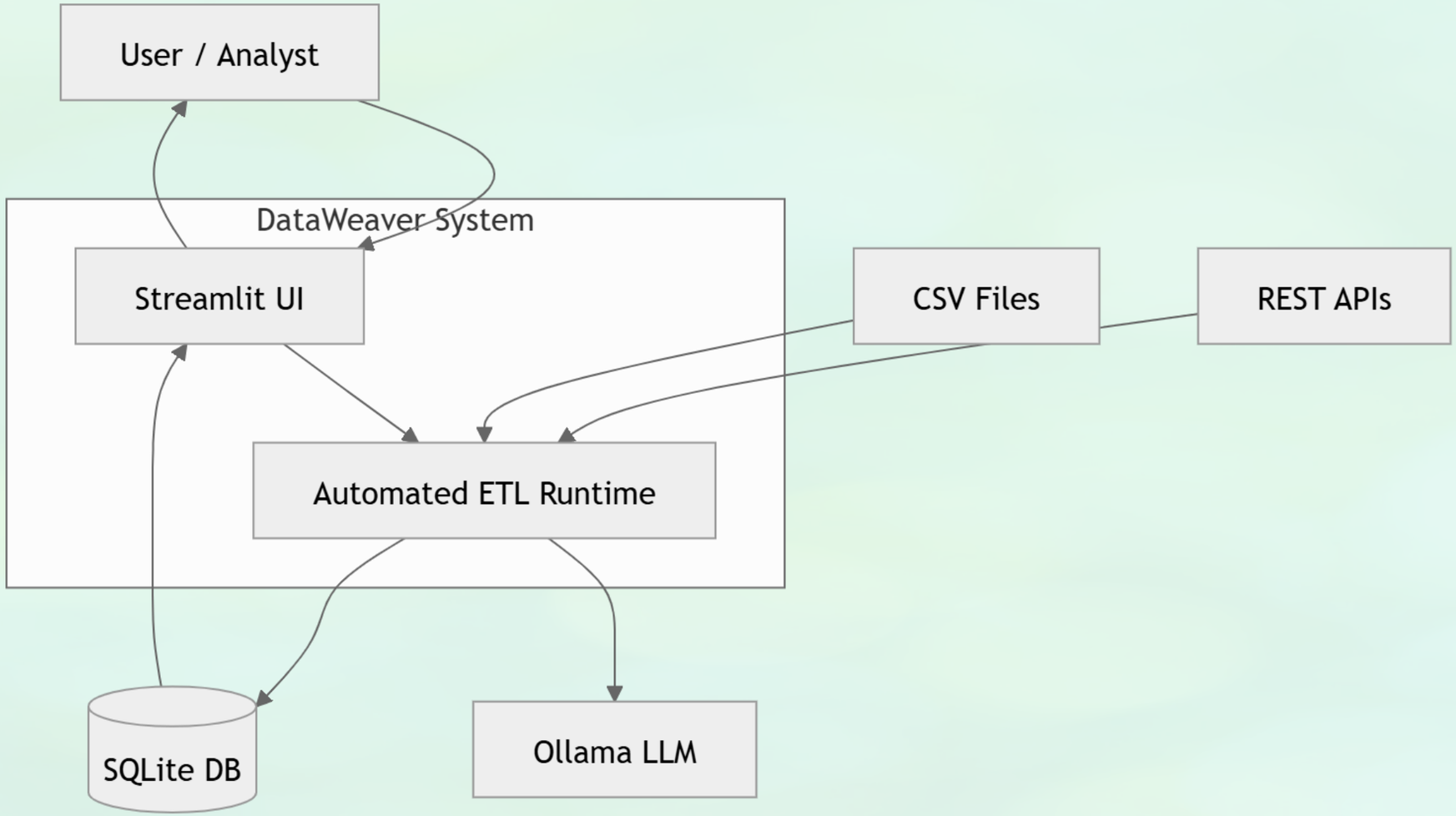
- **Data Processing:** Polars / Pandas (GPU-accelerated when available)

Architecture Diagram

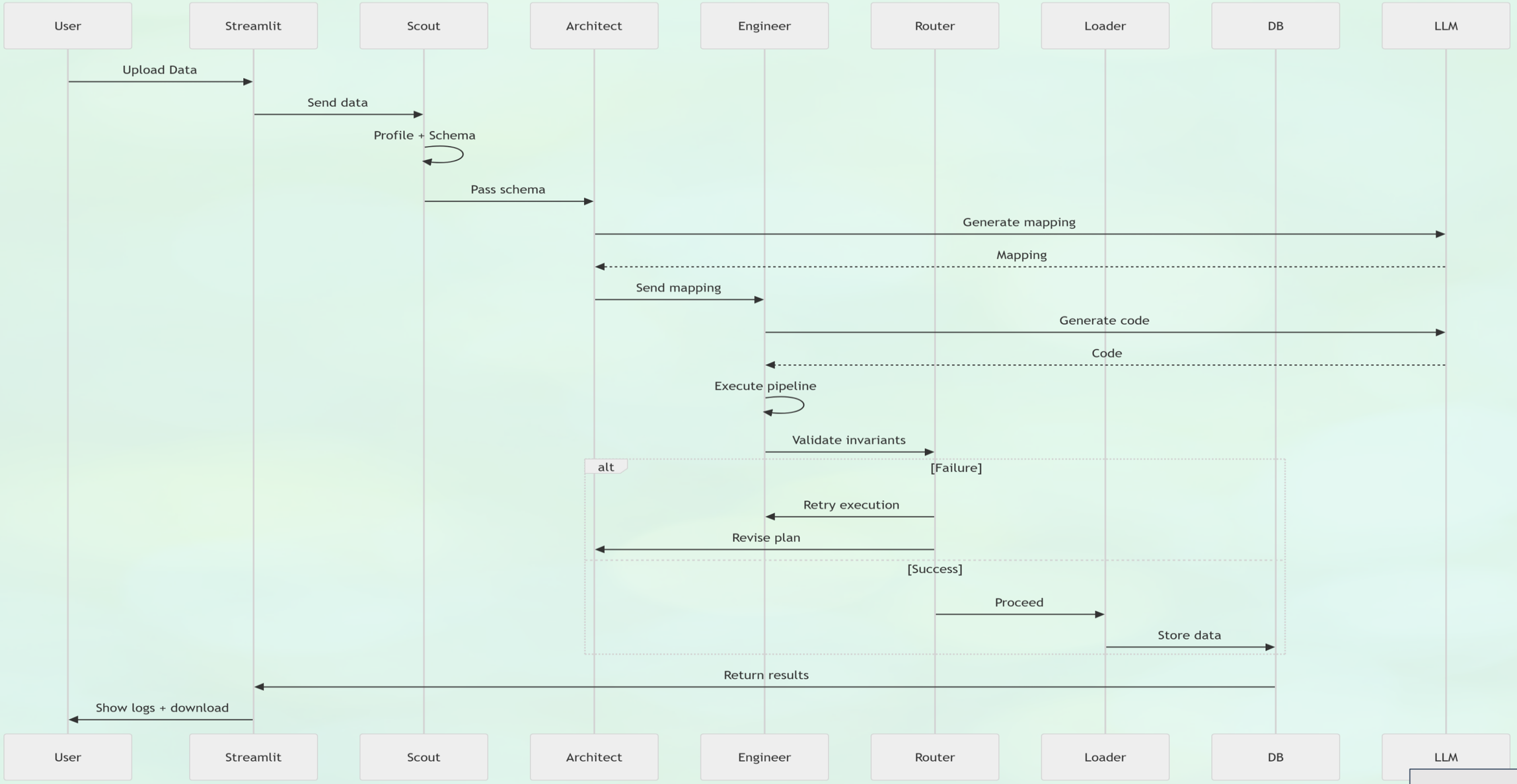


Sprint 2

Context diagram

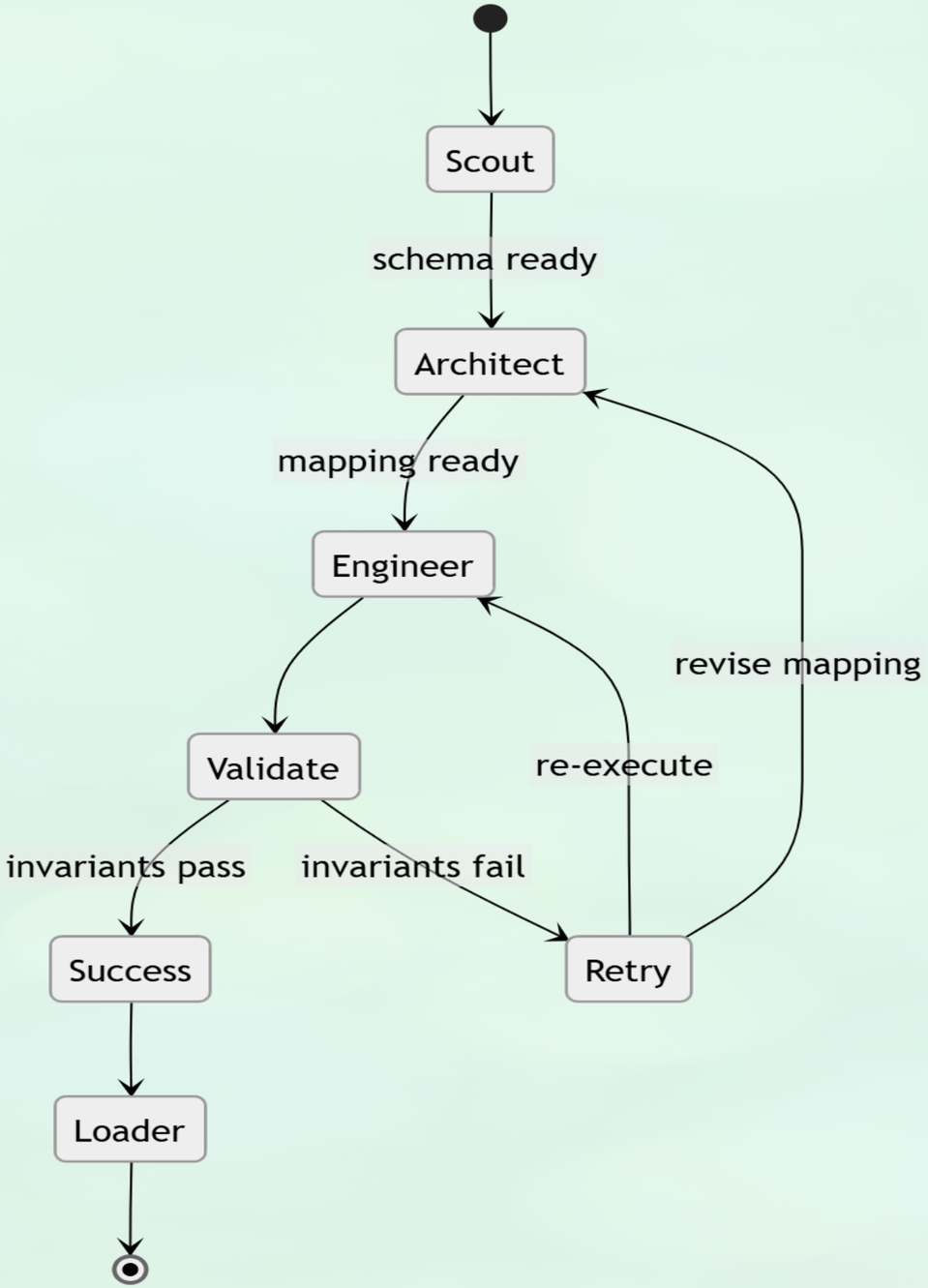


Sequence Diagram



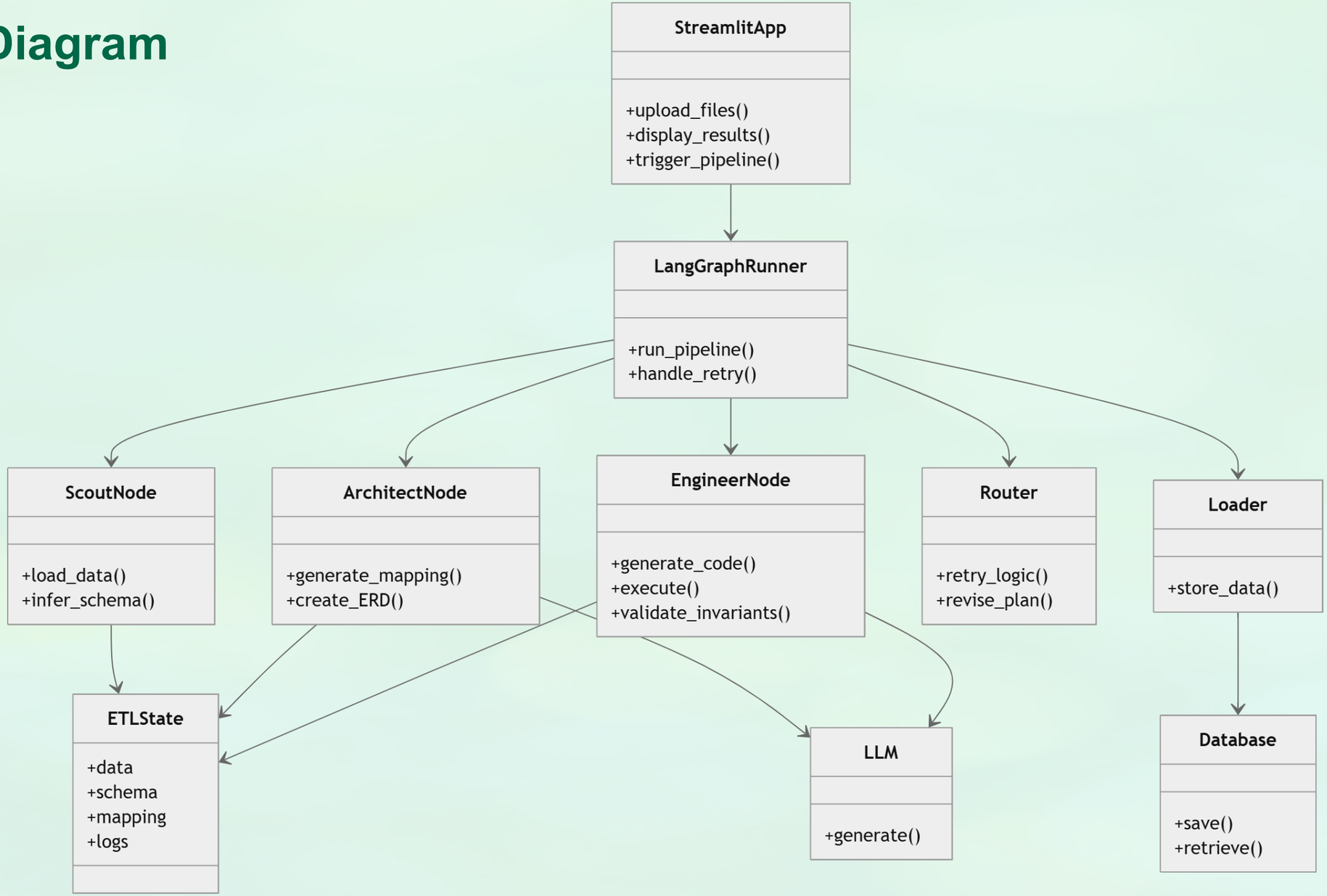
Sprint 2

State Diagram



Sprint 2

Class Diagram



Sprint 2

Sprint 1 - Recap 1

Slide 1: Data Ingestion & Validation

Sprint Goal: Build core pipeline for data upload and validation

Completed User Stories

- **US_001 – Data Upload**

  - CSV upload with format validation & confirmation

- **US_002 – Data Validation**

  - Detects missing/invalid fields with clear errors

Key Outcomes:

- Established reliable data ingestion layer
- Reduced early-stage data issues through validation
- Improved user confidence in data quality

Sprint 1 - Recap 1

Slide 2: Data Modeling & Transformation

Completed User Stories

- **US_003 – Schema Mapping**

  - PK–FK detection, column mapping (cust_id ↔ user_id)

- **US_004 – ERD Visualization**

  - Auto-generated, interactive ERD diagrams

- **US_005 – Data Transformation**

  - Joins, duplicate removal, schema casting

Key Outcomes:

- Enabled data relationship discovery
- Delivered visual understanding via ERD
- Produced clean, usable datasets

Product Backlog

1	ID	Assigned	Name	Type	Creation Date	US	Size	Priority	Dependencies
2	US1	Orges/Tenzing	API & CSV Tooling	F	2025-03-11	As a data engineer, I want the agent to have tools to connect to APIs and read CSV files, so that I can start extraction.	Small	High	None
3	US2	Orges/Tenzing	Data Extraction & Transformation	F	2025-03-11	As a data engineer, I want the agent to extract and transform data from APIs and CSVs, so I can integrate sources.	Small	High	US1
4	US3	Orges/Tenzing	Sensitive Data Redaction	F	2025-03-11	As a compliance officer, I want the agent to redact sensitive data, so we ensure privacy.	Medium	Medium	US2
5	US4	Orges/Tenzing	ETL Step Verification	F	2025-03-11	As a data steward, I want to verify each ETL step, so we ensure quality.	Small	High	US2
6	US5	Orges/Tenzing	Agent Coordination Framework	F	2025-03-11	As a team lead, I want a coordination framework, so agents collaborate on tasks.	Medium	Medium	None
7	US6	Orges/Tenzing	Parallel Multi-Agent Execution	F	2025-03-11	As a team lead, I want multiple agents to collaborate in parallel, so complex tasks finish faster.	Large	Medium	US5
8	US7	Shreya/Kush	Schema Change Monitoring	F	2025-03-11	As a data steward, I want to monitor dataset headers, so we catch schema changes.	Medium	Medium	US2
9	US8	Shreya/Kush	Pre-Join Data Sampling	F	2025-03-11	As a product owner, I want the agent to sample data before joining, so only clean data is used.	Medium	Medium	US2
10	US9	Orges/Tenzing	Conversational Memory	F	2025-03-11	As a user, I want the agent to remember past requests, so interactions are contextual.	Large	Low	Later enhancement
11	US9.1	Priya/Meghana	Pipeline Performance	NF	2025-03-11	As a data engineer, I want the ETL pipeline to complete a standard extraction and transformation job within a reasonable time threshold, so that the system is practical for real workloads and not just demos.	Medium	Medium	US1

Sprint 2

Product Backlog

12	US10	Priya/Meghana	Fault Tolerance	NF	2025-03-11	As a data engineer, I want the pipeline to gracefully handle agent failures without crashing the entire run, so that a single broken step doesn't require restarting the whole job from scratch.	Medium	High	US1
13	US11	ongoing	Credential & API Key Security	NF	2025-03-11	As a compliance officer, I want API keys and database credentials to be stored securely and never logged in plain text, so that sensitive configuration data is not exposed in traces or outputs.	Small	High	US1
14	US12	ongoing	Model Resource Allocation	NF	2025-03-11	As a data engineer, I want LLM-based agents to be assigned to the appropriate GPU based on model size and VRAM requirements, so that the pipeline runs efficiently without memory errors.	Small	High	None
15	US13	Priya/Meghana	Pipeline Maintainability	NF	2025-03-11	As a developer, I want each agent to be modular and independently testable, so that changes to one agent don't require refactoring the entire pipeline.	Medium	Medium	None
16	US14	ongoing	Traceability & Auditability	NF	2025-03-11	As a data steward, I want every agent action and decision to be logged with a timestamp and input/output summary, so that the pipeline can be audited after a run completes.	Small	High	US6
17	US15	sprint 3	Scalability	NF	2025-03-11	As a data engineer, I want the pipeline to handle datasets of increasing size without significant performance degradation, so that it remains viable beyond small demo datasets.	Large	Low	Later enhancement
18	US16	Shreya/Kush	Error Detection During Extraction/Transformation	F	2026-03-11	As a user, I want the system to detect and notify errors during data extraction or transformation so that I can correct issues before processing continues.	Medium	High	Data processing engine, validation rules

Sprint 2

Product Backlog

19	US17	Shreya/Kush	Multi-Format Output Export	F	2026-03-11	As a user, I want to export processed data in formats like CSV, Excel, Google Sheets, and JSON so that I can use the data in other tools.	Medium	High	File export service, Google Sheets API
20	US18	Shreya/Kush	Drag & Drop / API / Link Upload	F	2026-03-11	As a user, I want to upload files via drag & drop or provide APIs/links in a textbox so that I can easily ingest data from different sources.	Large	High	uploader, API connector service
21	US19	Shreya/Kush	Database Preview Before & After Processing	F	2026-03-11	As a user, I want to preview the database before and after processing in table or ERD format so that I can understand the structure and transformations applied.	Large	Medium	Database visualization library, transformation engine
22	US20	Shreya/Kush	Basic Analytics Functions	F	2026-03-11	As a user, I want basic analytics like average, median, and summary stats so that I can quickly analyze my dataset without external tools.	Medium	Medium	Data processing engine, aggregation functions
23	US21	Shreya/Kush	Dropdown Filters for Data	F	2026-03-11	As a user, I want dropdown filtering options so that I can easily filter datasets and focus on relevant information.	Small	Medium	UI filtering components, dataset query layer
24	US22	ongoing	Data Quality Validation	F	2026-03-11	As a user, I want the system to validate dataset quality so that I can detect missing values, duplicates, and invalid data before processing. **Acceptance Criteria** Detect missing values Detect duplicate rows Validate column data types Generate validation summary	Medium	Medium	
25	US23	sprint 3	Dataset Health Score	NF	2026-03-11	As a user, I want the system to generate a dataset health score so that I can quickly understand the overall quality of my dataset. **Acceptance Criteria** Health score calculated from validation metrics Score range from 0–100 Health classification displayed (Healthy, Moderate, Poor) Score displayed in dataset report	Small	Medium	

Sprint 2

Product Backlog

26	US24	Priya/Meghana	Agent Decision Timeline	F	2026-03-11	that I can understand how the system processed my dataset. **Acceptance Criteria** Each agent action is logged Timeline shows sequence of agent operations Each step displays agent name and action Timeline available in final dataset report	Small	Medium	
27	US25	Shreya/Kush	Dataset Profiling	F	2026-03-11	provide summary statistics so that I can quickly understand the structure and distribution of my data. **Acceptance Criteria** System detects number of rows and columns System identifies column data types System calculates summary statistics for numeric columns System identifies categorical column distributions Profiling report generated	Medium	Medium	
28	US26	sprint 3	Sandbox Dry-Run Mode	F	13-03-2026	As a data engineer, I want the agent to support a sandbox dry-run mode on a small sample of the data so that I can see all planned transformations and health checks before committing changes to the full dataset.	Medium	Medium	
29	US27	sprint 3	Auto-Generated Run Docs	F	13-03-2026	As a product owner, I want the agent to generate human-readable documentation for each ETL run so that stakeholders can quickly understand what inputs, transformations, and outputs were involved.	Small	Medium	US26,US25
30	US28	sprint 3	User Feedback & Rating Loop	F	13-03-2026	As an end user, I want to rate pipeline runs and leave feedback so that the agent can suggest improvements to mappings, validations, or configurations over time.	Small	Medium	US27

Sprint 2

Sprint 2 Backlog

1	No.	Description	Acceptance Criteria	Feature	Story points	Priority	Assignee
2	US_S2_01	As a data steward, I want to validate and monitor data quality across the ETL pipeline so that errors are detected early and data integrity is maintained.	* ETL step execution logs are captured for each stage * Row-level errors are detected and clearly reported * Schema validation ensures input and output consistency * Null values and anomalies are identified correctly * Validation reports summarize all detected issues * API returns validation results in structured format * System prevents invalid data from progressing in pipeline	Data Validation & Quality	55	High	Tenzing / Orges
3	US_S2_02	As a system, I want agents to coordinate and execute tasks in parallel so that ETL performance and scalability improve.	* ETLState contains all required shared fields * Agents execute tasks in correct workflow order * Parallel execution runs without data conflicts * Retry mechanism works correctly for failed steps * Agent logs capture execution details and decisions * Pipeline execution completes successfully under coordination * System handles partial failures without crashing entire flow	Multi-Agent System	55	Medium	Tenzing / Orges
4	US_S2_03	As a user, I want flexible data ingestion methods so that I can easily upload datasets from multiple sources.	* File upload supports drag & drop functionality * API-based upload accepts valid file formats (CSV, JSON) * Upload progress is displayed to the user * Invalid files are rejected with clear error messages * Uploaded datasets are stored successfully * System handles large file uploads without failure	Data Input	34	High	Shreya / Kushawnth
5	US_S2_04	As a user, I want to export processed data in multiple formats so that I can use it externally.	* Processed data can be exported in CSV format * Processed data can be exported in JSON format * Processed data can be exported in Excel format * Download API returns correct file response * Exported data maintains correct schema and formatting * System handles export errors gracefully	Data Output	34	High	Tenzing / Orges

Sprint 2

Sprint 2 Backlog

6	US_S2_05	As a user, I want to preview data before and after processing so that I understand transformations clearly.	* Dataset preview displays tabular data correctly * Schema view shows column names and data types * ERD visualization represents dataset relationships * Before and after data comparison is available * UI loads data efficiently without performance issues * API returns preview data accurately	Data Visualization	34	Medium	Shreya / Kushawnth
7	US_S2_06	As a user, I want basic analytics and filtering tools so that I can quickly analyze datasets.	* System calculates average and median correctly * Summary statistics are generated for datasets * Backend supports filtering queries * Aggregation results are returned in structured format * Queries execute efficiently on large datasets * Errors are handled and reported clearly	Analytics	34	Medium	Tenzing / Orges
8	US_S2_07	As a compliance officer, I want sensitive data to be redacted so that user privacy is protected.	*Schema changes are detected between pipeline runs *System identifies added, removed, or modified fields *Alerts are generated for unexpected schema changes *Schema change logs are stored for auditing *Monitoring integrates with ETL pipeline execution	Security	21	Medium	Tenzing / Orges

Sprint 2 Stories Completed

No.	User Story	Completed Date	Story point estimate	Completed
US_S2_01	**As a data steward, I want to validate and monitor data quality across the ETL pipeline so that errors are detected early and data integrity is maintained.**		**55**	**49**
	Implement ETL step validation logging	31-Mar	8	8
	Detect row-level errors during transformation	01-Apr	13	13
	Implement schema change monitoring	02-Apr	8	8
	Add null / anomaly detection checks	03-Apr	8	8
	Build validation report generation	04-Apr	8	6
	API endpoint to fetch validation results	05-Apr	5	**4**
	Error handling improvements	06-Apr	5	**2**
US_S2_02	**As a system, I want agents to coordinate and execute tasks in parallel so that ETL performance and scalability improve.**		**55**	**47**
	Define ETLState structure (shared state)	07-Apr	8	8
	Implement agent coordination logic	08-Apr	8	8
	Enable parallel execution of agents	09-Apr	13	13
	Add retry and failure handling	10-Apr	8	6
	Logging for agent decisions	11-Apr	5	**4**
	Performance benchmarking	12-Apr	8	**6**
	Execution monitoring hooks	13-Apr	5	**2**
US_S2_03	**As a user, I want to upload files via drag and drop or API so that I can easily ingest data from different sources.**		**34**	**31**
	Implement drag & drop file upload UI	14-Apr	8	8
	Add API-based upload support	15-Apr	8	7
	Upload progress indicator	16-Apr	5	5
	File selection UI improvements	16-Apr	5	5
	UX improvements	16-Apr	4	**3**
	Error display messages	16-Apr	4	**3**

Sprint 2 Stories Completed

US_S2_04	**As a user, I want to export processed data in multiple formats so that I can use it across different tools and platforms.**		**34**	**28**
	Implement CSV export	14-Apr	8	8
	Implement JSON export	14-Apr	5	5
	Implement Excel export	15-Apr	8	6
	Add download API endpoint	15-Apr	5	**4**
	Ensure consistent formatting	16-Apr	4	**3**
	Export error handling	16-Apr	4	**2**
US_S2_05	**As a user, I want to preview the dataset and understand schema changes before and after processing so that I can validate transformations visually.**		**34**	**30**
	Display dataset table preview	14-Apr	5	5
	Implement schema view	14-Apr	5	5
	Generate ERD visualization	15-Apr	8	**7**
	Show before vs after comparison	15-Apr	8	**7**
	Connect backend API to UI	16-Apr	4	**3**
	Optimize large dataset rendering	16-Apr	4	**3**
US_S2_06	**As a user, I want basic analytics and filtering capabilities so that I can quickly analyze and explore dataset insights.**		**34**	**29**
	Implement average, median calculations	14-Apr	5	5
	Generate summary statistics	14-Apr	5	5
	Query handling logic	15-Apr	8	**7**
	Filter query backend support	15-Apr	8	**6**
	Data aggregation optimization	16-Apr	4	**3**
	Error handling	16-Apr	4	**3**
US_S2_07	**As a data steward, I want to detect schema changes and receive alerts so that I can ensure data consistency and integrity across the pipeline.**		**21**	**18**
	Schema comparison logic	14-Apr	5	5
	Change detection rules	14-Apr	4	4
	Alert generation	15-Apr	4	**3**
	Logging changes	15-Apr	4	**3**
	Integration with pipeline	16-Apr	4	**3**

Sprint 2 Test cases

1	Module	Test Case / Scenario	Test Steps	Expected Result	Priority	Status
2	Data Validation	Validate row-count invariant after join	Upload two datasets; Perform join; Check row count	Output rows ≤ product of input rows	High	Pass
3	Data Validation	Detect schema mismatch	Upload dataset with incorrect schema; Run pipeline	System flags schema inconsistency	High	Pass
4	Data Validation	Detect null spike	Upload dataset with missing values; Run transformation	System flags abnormal null increase	Medium	Fail
5	Agent Coordination	Verify Mapper and Aggregator interaction	Upload dataset; Trigger pipeline	Mapper generates schema, Aggregator executes	High	Pass
6	Agent Coordination	Validate error handling between agents	Introduce invalid schema; Run pipeline	Aggregator halts and reports error	High	Pass
7	Agent Coordination	Check communication latency	Run large dataset; Monitor logs	Agents coordinate without failure	Medium	Pass
8	File Upload	Upload valid CSV file	Drag and drop valid CSV	File uploads successfully	High	Pass
9	File Upload	Upload invalid file format	Upload .txt file	System rejects file with error	High	Pass
10	File Upload	Upload large file	Upload large dataset	File uploads without crash	Medium	Pass
11	Data Export	Export processed dataset	Run pipeline; Click export	File downloads successfully	High	Pass
12	Data Export	Verify export format	Export dataset	File matches expected format (CSV/JSON)	High	Pass
13	Data Export	Export empty dataset	Run pipeline with no output	System shows warning	Medium	Fail
14	Data Preview	Preview dataset	Upload dataset; View preview	Data displayed correctly	High	Pass
15	Data Preview	Handle large dataset preview	Upload large dataset	Preview loads with pagination	Medium	Pass
16	Data Preview	Validate column metadata display	Upload dataset	Column names/types displayed	Medium	Pass
17	Analytics	Apply filter on dataset	Upload dataset; Apply filter	Filtered results displayed	High	Pass
18	Analytics	Generate summary stats	Upload dataset; Run analytics	Stats (mean, count, etc.) shown	High	Pass
19	Analytics	Invalid filter handling	Apply invalid filter	System shows error	Medium	Pass
20	Schema Detection	Detect schema changes	Upload updated dataset	System flags schema difference	High	Pass
21	Schema Detection	Handle compatible schema changes	Upload slightly modified schema	System adapts without failure	Medium	Pass
22	Schema Detection	Log schema change history	Upload multiple versions	System logs changes	Low	Pass

Team Velocity

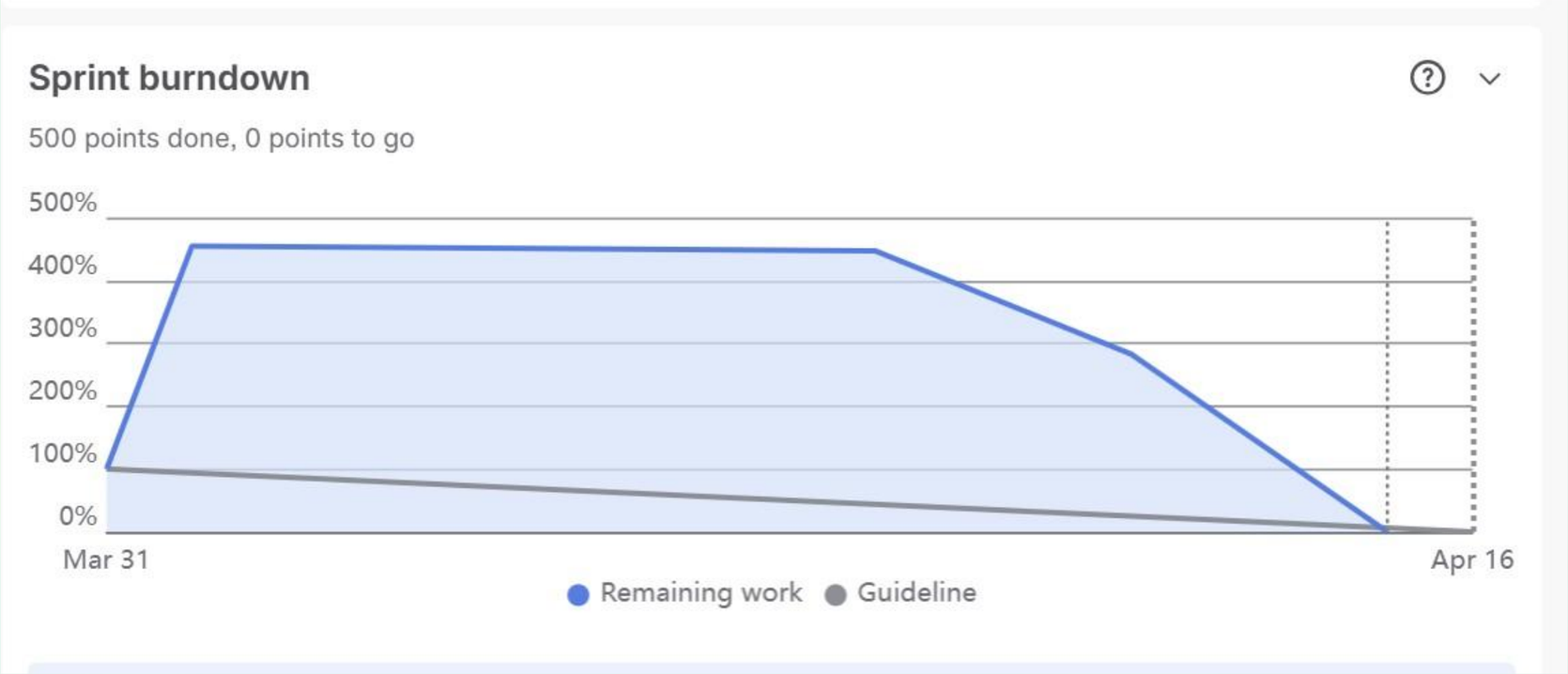
- Total estimated points:267
- Total points delivered: 232
- Total points carried over: 35
- Team Velocity: 232

WEAVE Sprint 2 31 Mar – 16 Apr (13 work items) | 0 | 0 | 236 | Complete sprint

~~WEAVE-174~~	Sprint Planning		DONE	-	
~~WEAVE-175~~	Create User Stories		DONE	-	PP
~~WEAVE-119~~	US_S2_01 — Data Quality & Validation Pipeline	DATA VALIDATION & ...	DONE	55	TS
~~WEAVE-120~~	US_S2_02 — Multi-Agent Coordination & Execution	MULTI-AGENT SYSTEM	DONE	55	OV
~~WEAVE-122~~	US_S2_03 — Data Ingestion & Upload Experience	DATA INPUT	DONE	34	TS
~~WEAVE-124~~	US_S2_04 — Data Output & Export System	DATA OUTPUT	DONE	34	OV
~~WEAVE-126~~	US_S2_05 — Data Preview & Visualization	DATA VISUALIZATION	DONE	34	SK
~~WEAVE-127~~	US_S2_06 — Basic Analytics & Filtering	ANALYTICS	DONE	3	TS
~~WEAVE-129~~	US_S2_07 — Schema Monitoring	SECURITY	DONE	21	OV
~~WEAVE-176~~	Test Case Documentation		DONE	-	SK
WEAVE-177	Sprint 2 Retrospective and Recording		TO DO	-	
WEAVE-178	Sprint 3 Planning and Recording		IN PROGRESS	-	
WEAVE-179	Sprint 2 Presentation and Recording		IN PROGRESS	-	

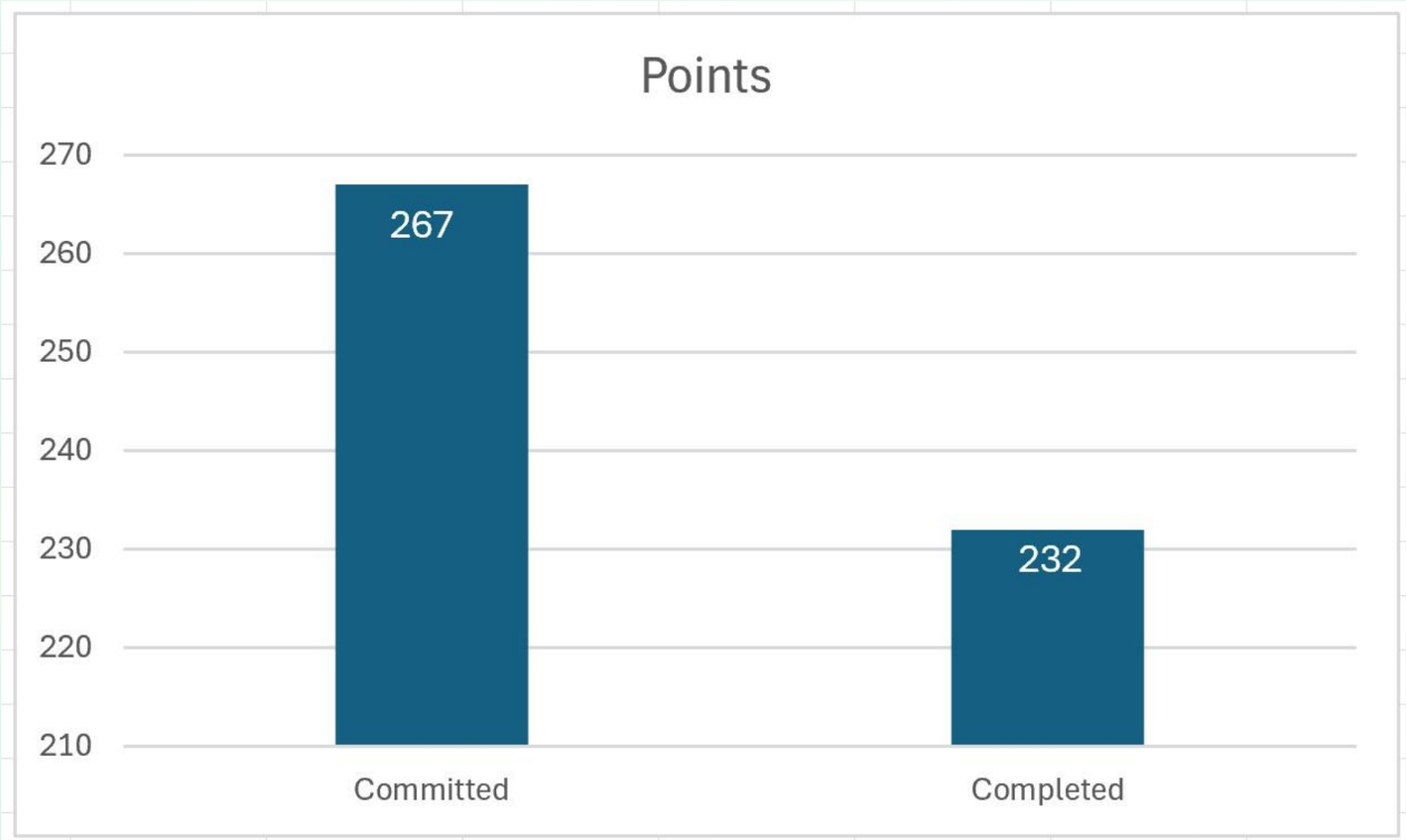
Sprint 2

Burndown Chart



Sprint 2

Completed/Committed Ratio



Sprint 2 Retrospective

- **What went well**
  - US_006 – Row Count Validation -Prevented incorrect joins
  - US_007 – Schema Validation - Ensured output format consistency
  - US_008 – Null Detection - Identified data quality issues early
  - US_009 – Pipeline Monitoring - Real-time execution tracking
  - US_010 – Data Download - Easy export of processed data
- **What to improve**
  - Performance on large datasets (US_013)
  - API error handling robustness (US_012)
  - Logging & failure traceability (US_014)
- **Action items**
  - Implement distributed processing
  - Strengthen error handling & logging
  - Improve API resilience
  - Expand Data Dictionary (US_011)

Sprint 3 Planning

ID	Assigned	Name	Type	Creation Date	US	Size	Priority	Dependencies	Notes
US9	Tenzing/Orges	Pipeline Performance	NF	11-03-2025	As a data engineer, I want the ETL pipeline to complete a standard extraction and transformation job within a reasonable time threshold, so that the system is practical for real workloads and not just demos.	Medium	Medium	US1	
US10	Shreya/Kush	Fault Tolerance	NF	11-03-2025	As a data engineer, I want the pipeline to gracefully handle agent failures without crashing the entire run, so that a single broken step doesn't require restarting the whole job from scratch.	Medium	High	US1	
US11	Tenzing/Orges	Credential & API Key Security	NF	11-03-2025	As a compliance officer, I want API keys and database credentials to be stored securely and never logged in plain text, so that sensitive configuration data is not exposed in traces or outputs.	Small	High	US1	
US12	Tenzing/Orges	Model Resource Allocation	NF	11-03-2025	As a data engineer, I want LLM-based agents to be assigned to the appropriate GPU based on model size and VRAM requirements, so that the pipeline runs efficiently without memory errors.	Small	High	None	Specific to home lab setup — 5070 (16GB) for Architect, 4000 series (12GB) for Engineer.
US13	Tenzing/Orges	Pipeline Maintainability	NF	11-03-2025	As a developer, I want each agent to be modular and independently testable, so that changes to one agent don't require refactoring the entire pipeline.	Medium	Medium	None	
US14	Tenzing/Orges & Shreya/Kush	Traceability & Auditability	NF	11-03-2025	As a data steward, I want every agent action and decision to be logged with a timestamp and input/output summary, so that the pipeline can be audited after a run completes.	Small	High	US6	Distinct from US6 — this is post-run audit logging for stakeholders, not developer tracing.

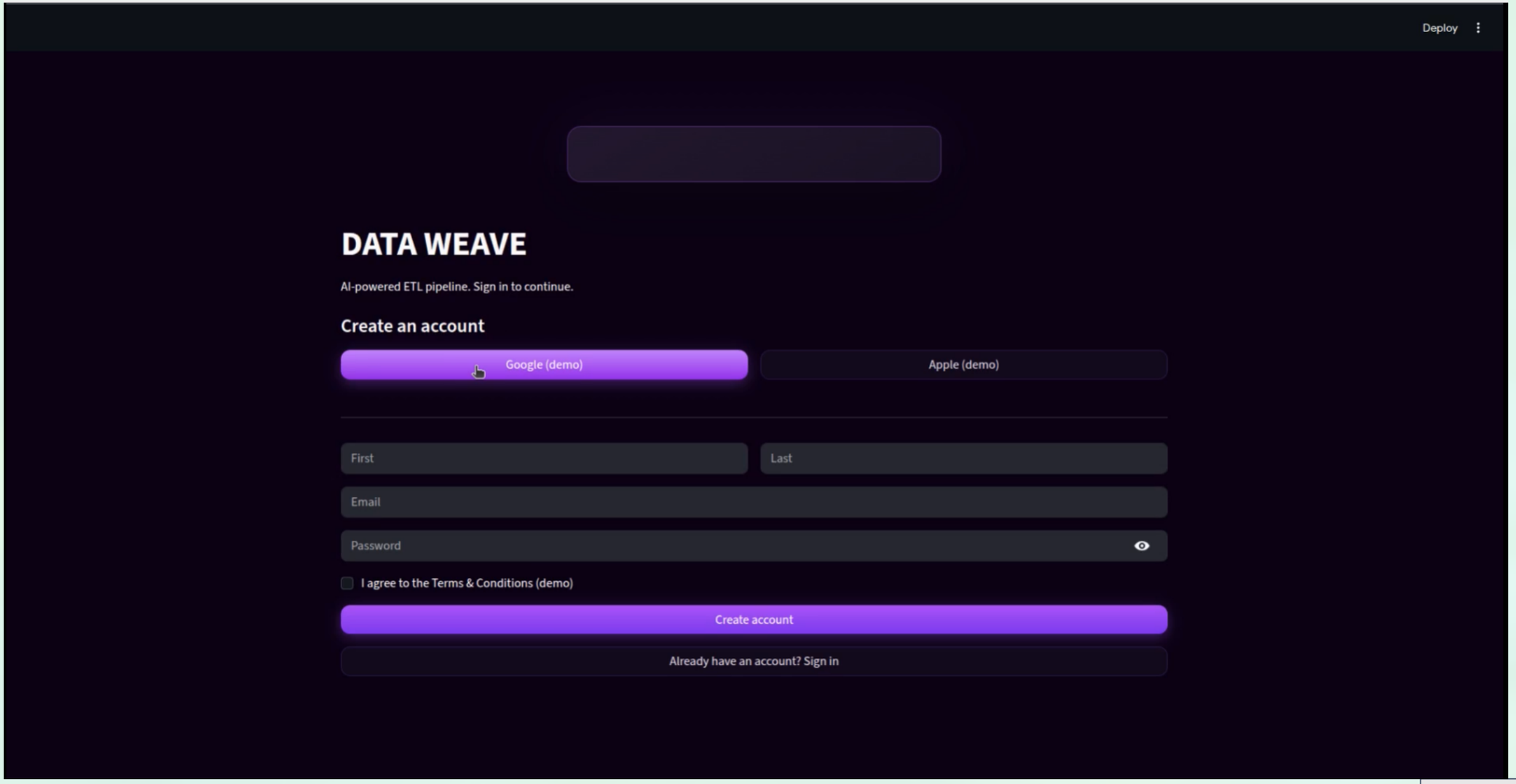
Sprint 2

Sprint 3 Planning

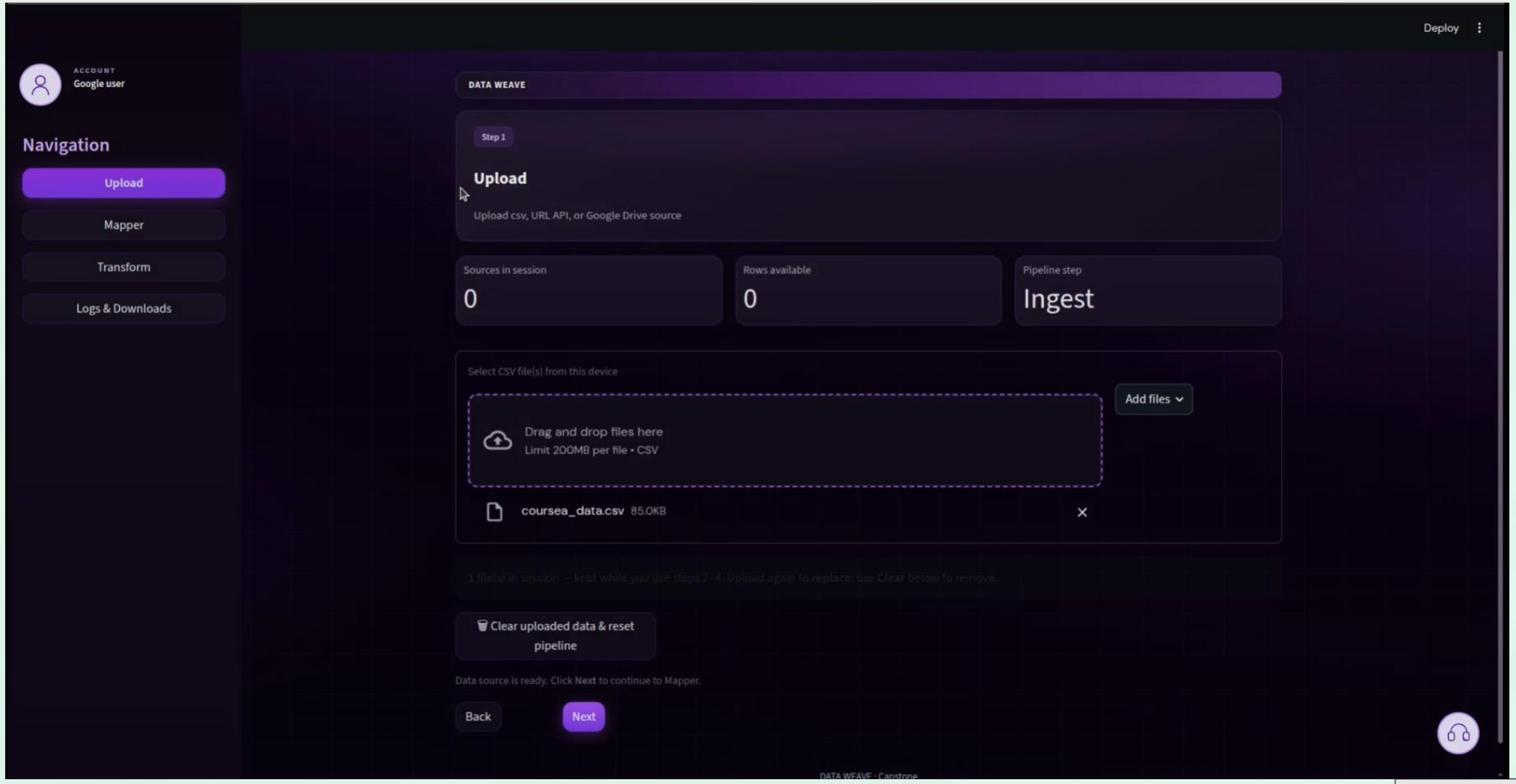
US15	Tenzing/orges	Scalability	NF	11-03-2025	As a data engineer, I want the pipeline to handle datasets of increasing size without significant performance degradation, so that it remains viable beyond small demo datasets.	Large	Low	Later enhancement
US22	Meghana/Priya	Data Quality Validation	F	11-03-2026	As a user, I want the system to validate dataset quality so that I can detect missing values, duplicates, and invalid data before processing. **Acceptance Criteria** Detect missing values Detect duplicate rows Validate column data types Generate validation summary	Medium	Medium	
US23	Meghana /Priya & Shreya/Kush	Dataset Health Score	NF	11-03-2026	As a user, I want the system to generate a dataset health score so that I can quickly understand the overall quality of my dataset. **Acceptance Criteria** Health score calculated from validation metrics Score range from 0–100 Health classification displayed (Healthy,Moderate, Poor) Score displayed in dataset report	Small	Medium	
US32	Shreya/Kush	User Feedback & Rating Loop	F	3/13/2026	As an end user, I want to rate pipeline runs and leave feedback so that the agent can suggest improvements to mappings, validations, or configurations over time.	Small	Medium	US27
US33	Tenzing/orges	Support for RDBMS	F	3/13/2026	User should be able to connect database to system	Medium	Medium	
US34	Tenzing/orges	External API support	F	3/13/2026	User should be able to connect API based dataset to system	Medium	Medium	

Made with GAMMA

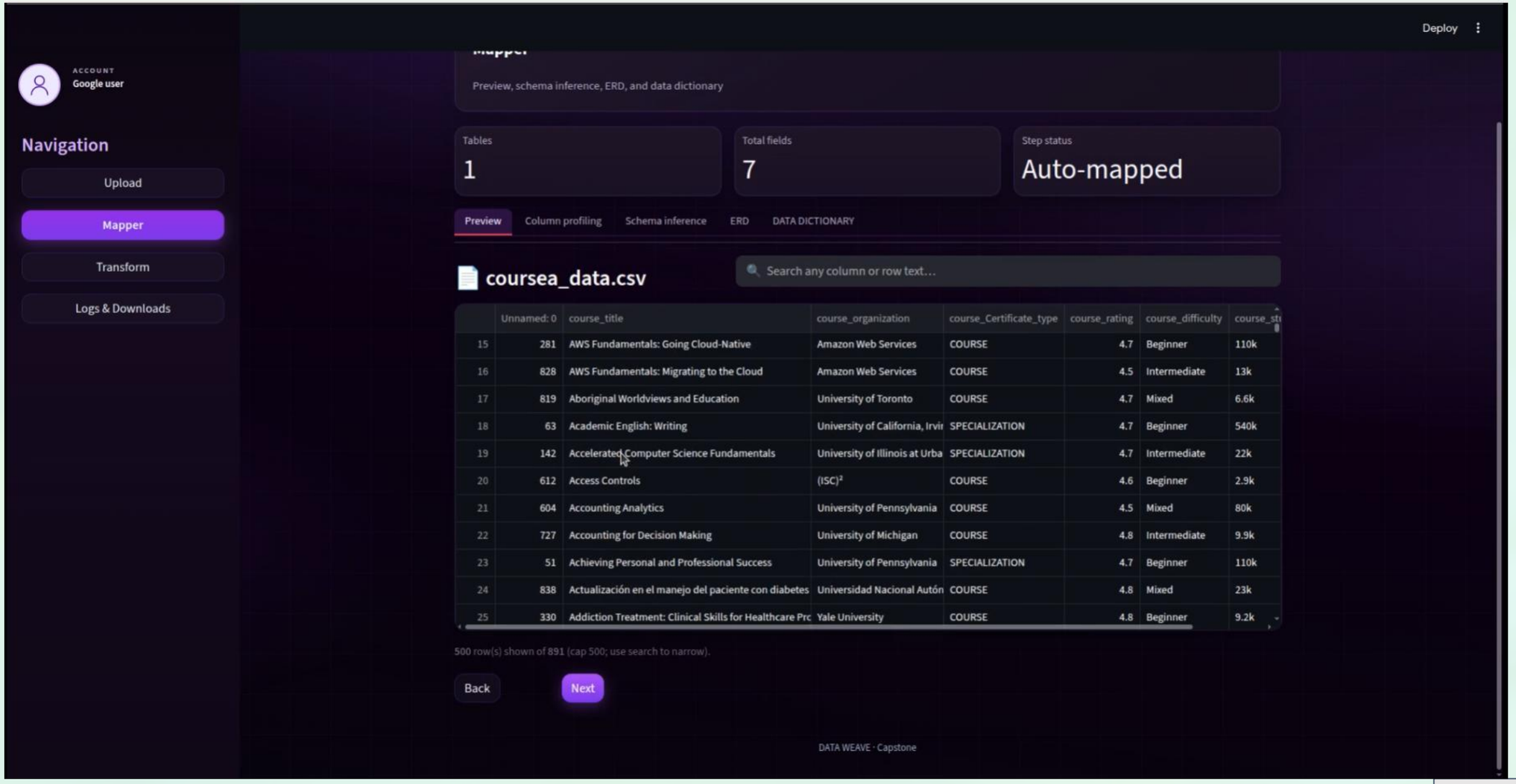
Sprint 2 Demo Slides Screenshots



Sprint 2 Demo Slides Screenshots

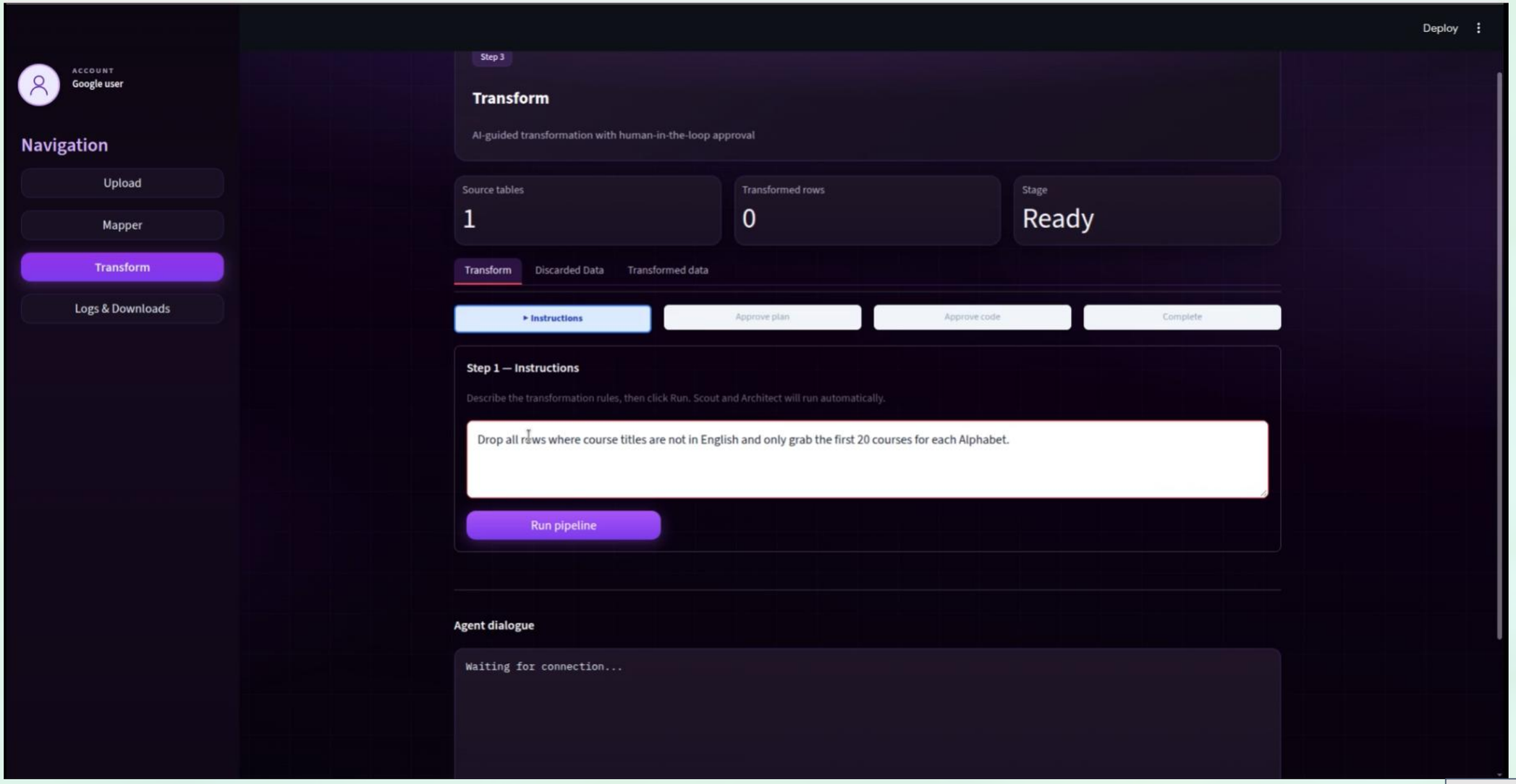


Sprint 2 Demo Slides Screenshots

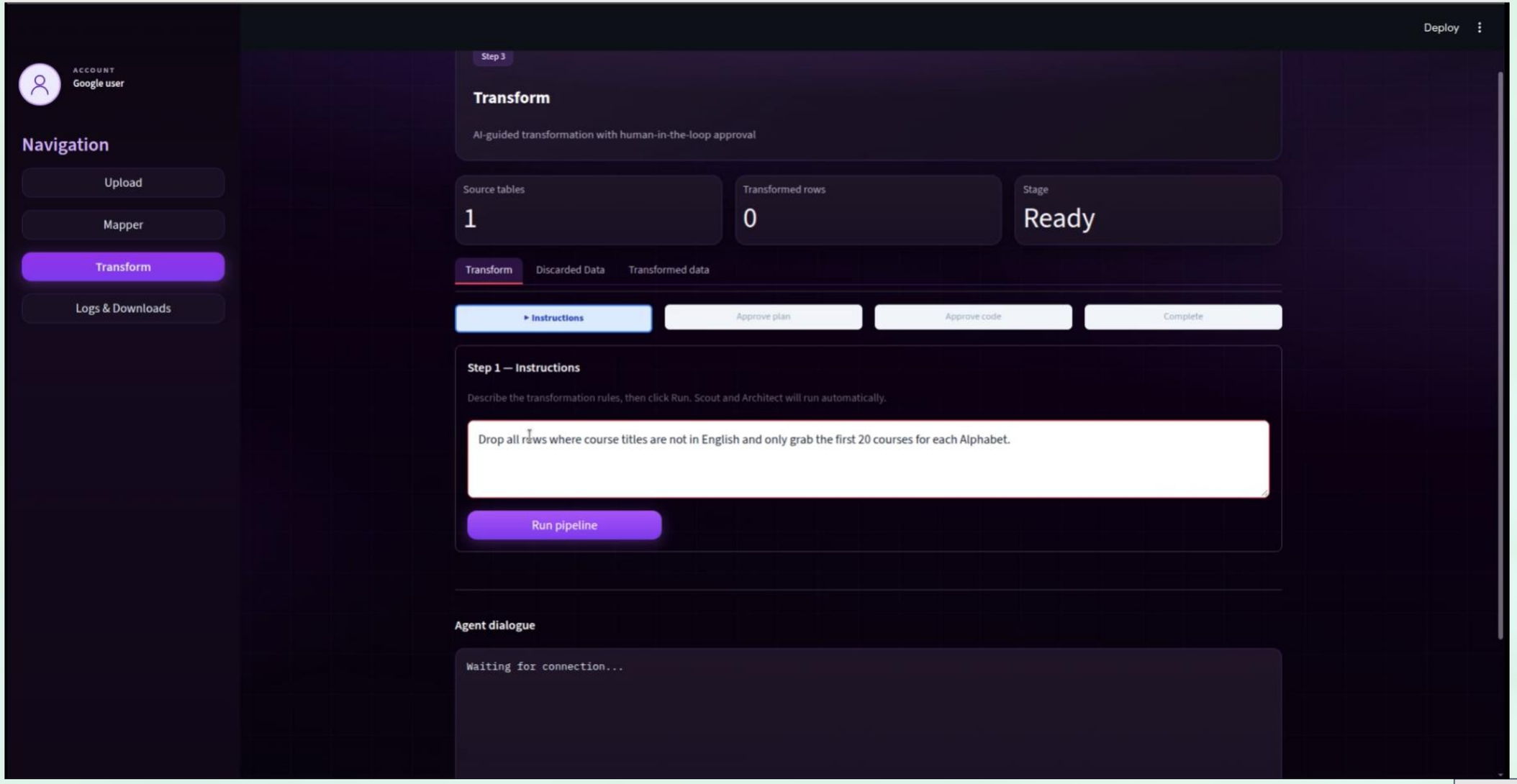


	Unnamed: 0	course_title	course_organization	course_Certificate_type	course_rating	course_difficulty	course_st
15	281	AWS Fundamentals: Going Cloud-Native	Amazon Web Services	COURSE	4.7	Beginner	110k
16	828	AWS Fundamentals: Migrating to the Cloud	Amazon Web Services	COURSE	4.5	Intermediate	13k
17	819	Aboriginal Worldviews and Education	University of Toronto	COURSE	4.7	Mixed	6.6k
18	63	Academic English: Writing	University of California, Irvi	SPECIALIZATION	4.7	Beginner	540k
19	142	Accelerated Computer Science Fundamentals	University of Illinois at Urba	SPECIALIZATION	4.7	Intermediate	22k
20	612	Access Controls	(ISC)²	COURSE	4.6	Beginner	2.9k
21	604	Accounting Analytics	University of Pennsylvania	COURSE	4.5	Mixed	80k
22	727	Accounting for Decision Making	University of Michigan	COURSE	4.8	Intermediate	9.9k
23	51	Achieving Personal and Professional Success	University of Pennsylvania	SPECIALIZATION	4.7	Beginner	110k
24	838	Actualización en el manejo del paciente con diabetes	Universidad Nacional Autón	COURSE	4.8	Mixed	23k
25	330	Addiction Treatment: Clinical Skills for Healthcare Prc	Yale University	COURSE	4.8	Beginner	9.2k

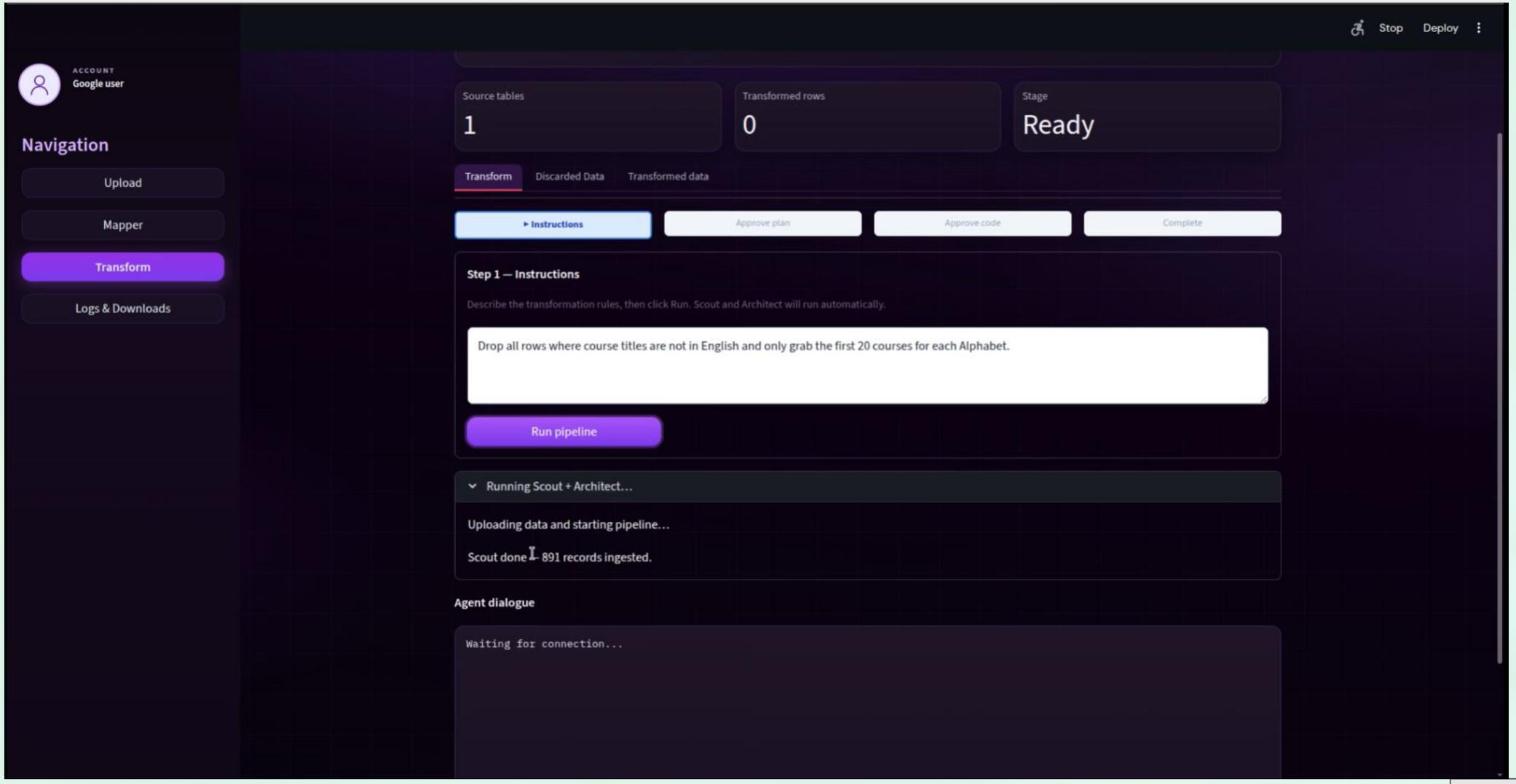
Sprint 2 Demo Slides Screenshots



Sprint 2 Demo Slides Screenshots



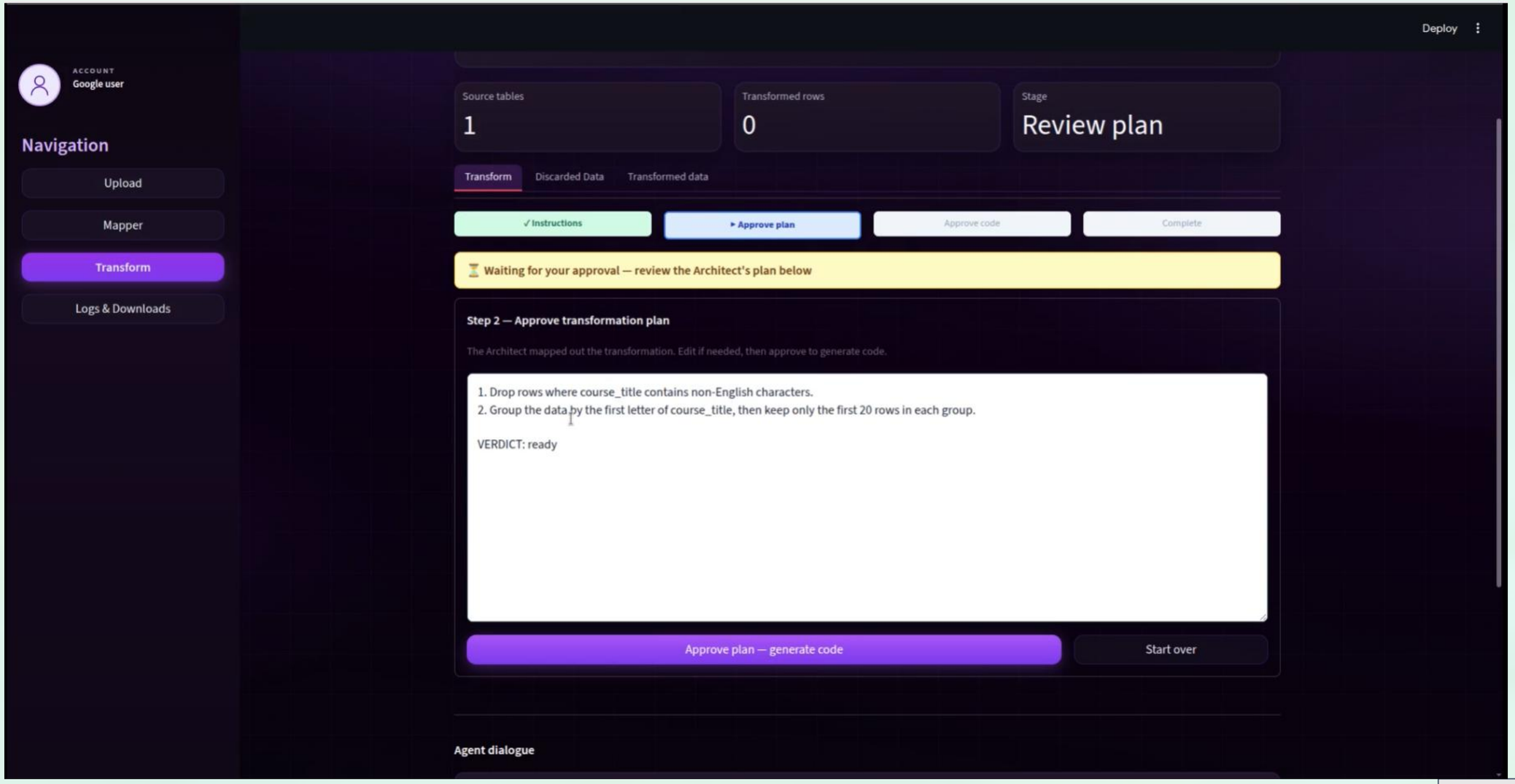
Sprint 2 Demo Slides Screenshots



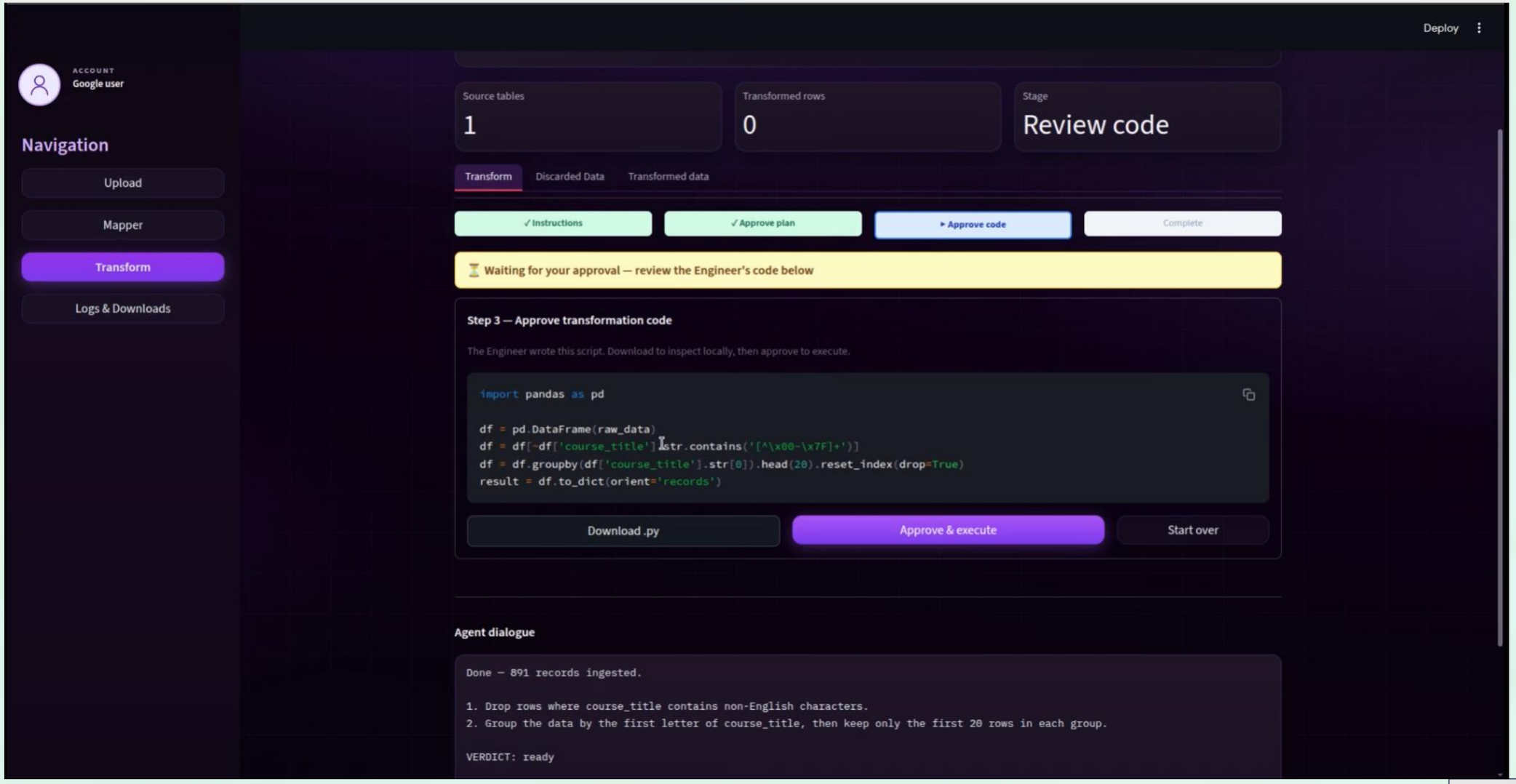
Sprint 2 Demo Slides Screenshots

```
Every 1.0s: nvidia-smi                                                                                  gaami: Thu Apr 16 14:20:21 2026

Thu Apr 16 14:20:21 2026
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 590.48.01              Driver Version: 590.48.01      CUDA Version: 13.1     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  NVIDIA GeForce RTX 5070 Ti     Off |   00000000:01:00.0  On |                  N/A |
| 30%   63C    P1            279W /  300W |   13399MiB /  16303MiB |     98%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+
|   1  NVIDIA GeForce RTX 3070        Off |   00000000:06:00.0 Off |                  N/A |
|  0%   50C    P2             50W /  270W |     326MiB /   8192MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI              PID   Type   Process name                        GPU Memory |
|        ID   ID                                                               Usage      |
|=========================================================================================|
|    0   N/A  N/A            6515      G   /usr/lib/xorg/Xorg                      340MiB |
|    0   N/A  N/A            6776      G   /usr/bin/gnome-shell                    128MiB |
|    0   N/A  N/A            7424      G   ...exec/xdg-desktop-portal-gnome          5MiB |
|    0   N/A  N/A            8280      G   ...rack-uuid=3190708988185955192         63MiB |
|    0   N/A  N/A           22432      G   /usr/share/code/code                     31MiB |
|    0   N/A  N/A           27830      G   /usr/bin/nautilus                        14MiB |
|    0   N/A  N/A           35591      G   /usr/share/discord/Discord               35MiB |
|    0   N/A  N/A           37015      G   .../8107/usr/lib/firefox/firefox         30MiB |
|    0   N/A  N/A           63418      C   /usr/local/bin/ollama                12612MiB |
|    1   N/A  N/A            6515      G   /usr/lib/xorg/Xorg                        4MiB |
|    1   N/A  N/A           60194      C   ...breoffice/program/soffice.bin        152MiB |
|    1   N/A  N/A           63418      C   /usr/local/bin/ollama                   154MiB |
+-----------------------------------------------------------------------------------------+
```

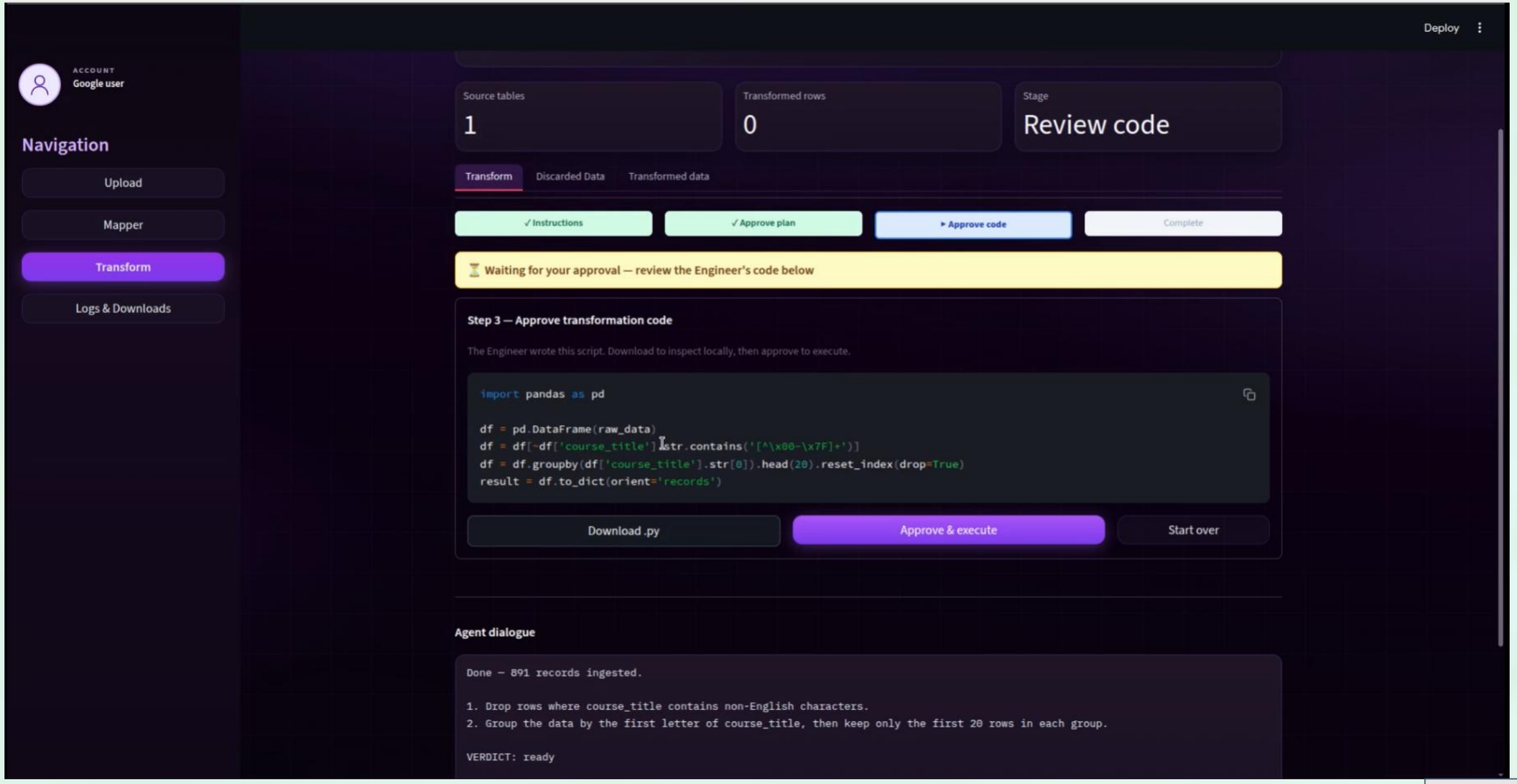
Sprint 2 Demo Slides Screenshots



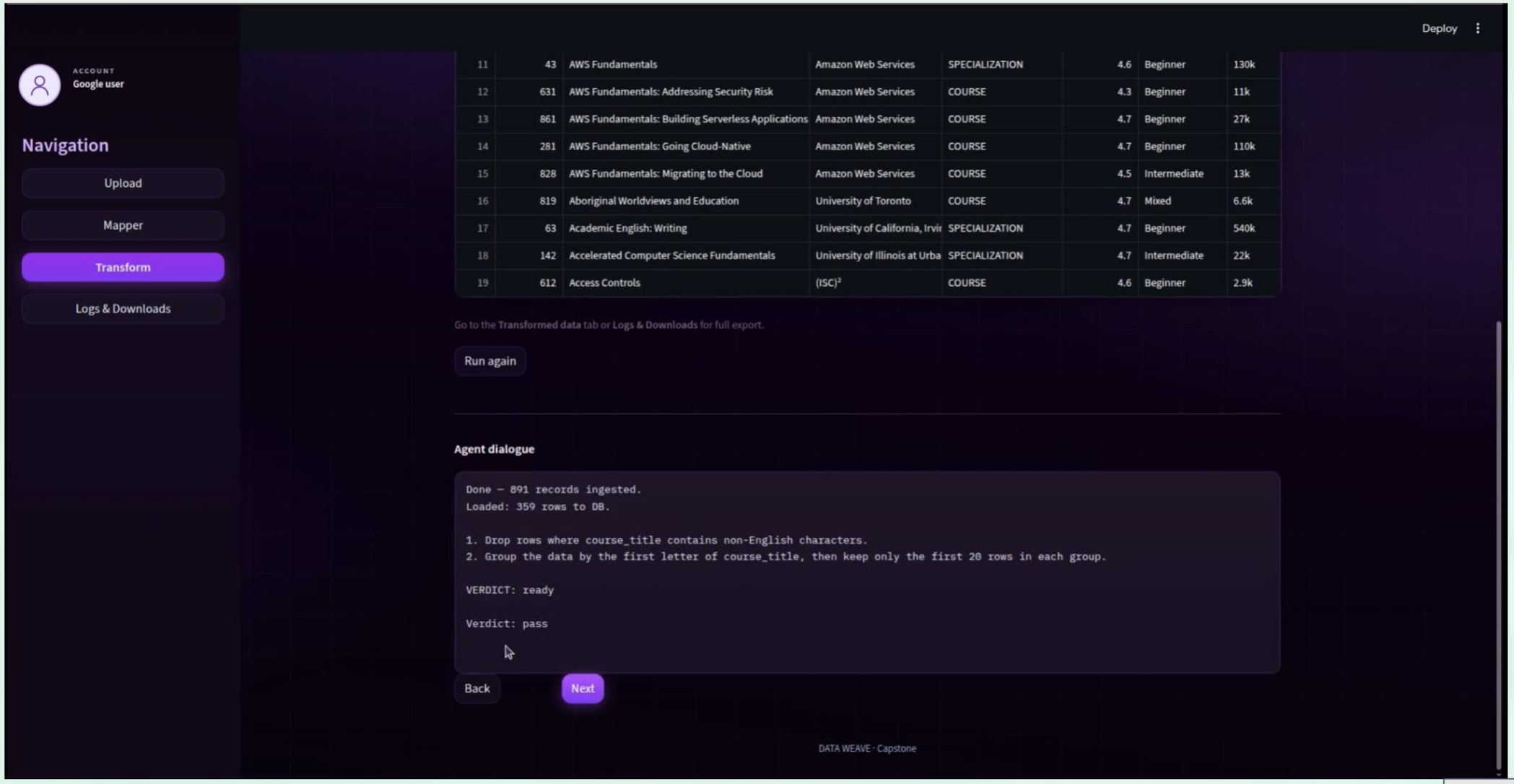
Sprint 2 Demo Slides Screenshots



Sprint 2 Demo Slides Screenshots

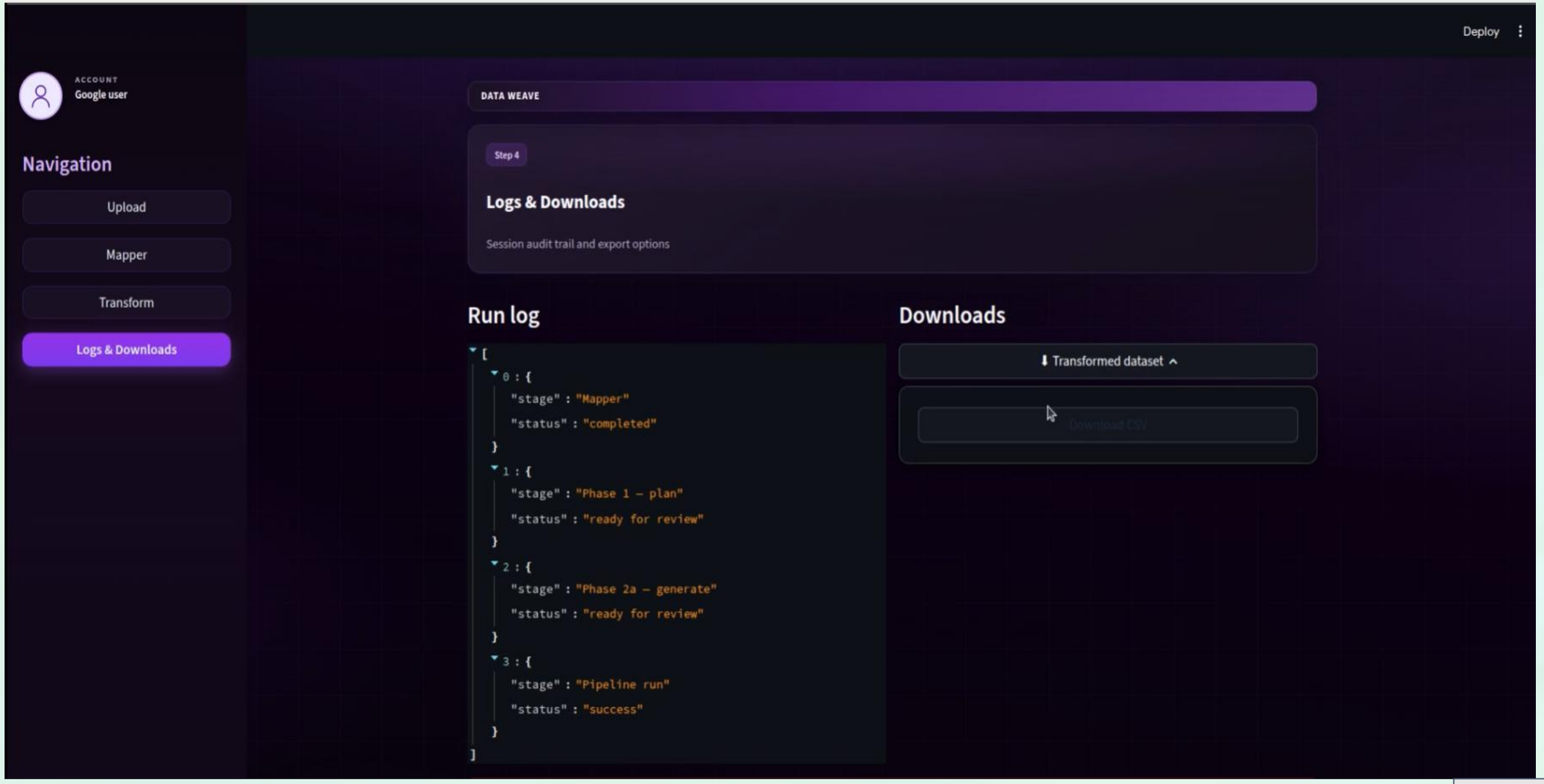


Sprint 2 Demo Slides Screenshots



11	43	AWS Fundamentals	Amazon Web Services	SPECIALIZATION	4.6	Beginner	130k
12	631	AWS Fundamentals: Addressing Security Risk	Amazon Web Services	COURSE	4.3	Beginner	11k
13	861	AWS Fundamentals: Building Serverless Applications	Amazon Web Services	COURSE	4.7	Beginner	27k
14	281	AWS Fundamentals: Going Cloud-Native	Amazon Web Services	COURSE	4.7	Beginner	110k
15	828	AWS Fundamentals: Migrating to the Cloud	Amazon Web Services	COURSE	4.5	Intermediate	13k
16	819	Aboriginal Worldviews and Education	University of Toronto	COURSE	4.7	Mixed	6.6k
17	63	Academic English: Writing	University of California, Irvir	SPECIALIZATION	4.7	Beginner	540k
18	142	Accelerated Computer Science Fundamentals	University of Illinois at Urba	SPECIALIZATION	4.7	Intermediate	22k
19	612	Access Controls	(ISC)²	COURSE	4.6	Beginner	2.9k

Sprint 2 Demo Slides Screenshots



THANK YOU

We sincerely appreciate your time and attention today

Wiki Page URL: https://github.com/htmw/S2026A-Team2/wiki